TidyTuesday Week 42

Pranay Gundam

Thursday 15th October, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we take two economic time series that have never met, have no business being in the same room, and force them into a linear regression together to see what happens. As always, the house motto applies: these pairs are chosen at random, no causal claim is being made, and spurious correlation isn't a bug here, it's basically the entire personality of the project. Please do not reallocate your retirement savings based on anything below.

Monday kicked things off with the Balance on Services (discontinued, naturally – all our best friends are discontinued) versus the exchange rate for Belize. The raw regression coughed up a p-value of 0.045, which is the statistical equivalent of squeaking through a door right as it closes. But here's the plot twist: the detrended version actually got stronger, flipping the sign to negative and tightening up to a p-value of 0.018. So after we stripped out the shared drift, these two STILL want to hang out, except now in opposite directions. I have no theory for why the US services balance would have anything to do with the Belizean dollar per US dollar, and I refuse to invent one, but credit where it's due: this little guy survived detrending, which is more than most contestants can say.

Then came Tuesday, and oh, it was a showstopper. Gasoline prices in the South Census Region versus 30-year FHA mortgage rates. The raw regression was significant, sure, but the detrended version is where things got genuinely spooky: an R-squared of 0.63 and a p-value with fifty-one zeros after the decimal point. Fifty-one! That's not a correlation, that's a marriage. And here's the uncomfortable part – this one actually makes a smidge of sense, because gas prices and mortgage rates both ride the same 1970s-to-1990s inflation rollercoaster, and that linkage held up even after we yanked out the trend. So this is the rare entry I'm legally obligated to call "actually kind of interesting," which pains me to type. Enjoy it, because it won't last.

Wednesday brought machinery corporations' cash piles versus government-sponsored enterprise debt, which in raw form posted a jaw-dropping R-squared of 0.80 – before immediately face-planting into irrelevance the moment we detrended it (p-value crashing to a limp 0.13). This is the textbook trend mirage: two things that both went up over time, holding hands only because the calendar was dragging them both forward. Classic. Thursday closed us out with Sweden's government consumption prices versus Kansas City energy costs, which looked respectable raw ($p = 0.0004$) and then evaporated completely once detrended, landing at a gloriously meaningless p-value of 0.67. Two for two on the mirages this week, which is honestly the natural order of things. See you next week, when we once again mistake coincidence for destiny.

2 Date: 2026-10-12

Series ID: BOPBSVN
This series is titled Balance on Services (DISCONTINUED) and has a frequency of Quarterly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1960-01-01 and the observation end date is 2014-01-01.The popularity of this series is 1.

Series ID: XRNCUSBZA618NRUG
This series is titled Exchange Rate (market+estimated) for Belize and has a frequency of Annual. The units are National Currency Units per US Dollar and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1970-01-01 and the observation end date is 2023-01-01.The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_XRNCUSBZA618NRUG	**R-squared:**	0.090
Model:	OLS	**Adj. R-squared:**	0.069
Method:	Least Squares	**F-statistic:**	4.246
Date:	Mon, 12 Oct 2026	**Prob (F-statistic):**	0.0454
Time:	20:28:58	**Log-Likelihood:**	34.026
No. Observations:	45	**AIC:**	-64.05
Df Residuals:	43	**BIC:**	-60.44
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.9274	0.024	79.937	0.000	1.879	1.976
value_fred_BOPBSVN	0.0023	0.001	2.060	0.045	4.83e-05	0.004

Omnibus:	17.591	**Durbin-Watson:**	0.425
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	23.370
Skew:	-1.275	**Prob(JB):**	8.42e-06
Kurtosis:	5.441	**Cond. No.**	30.5

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

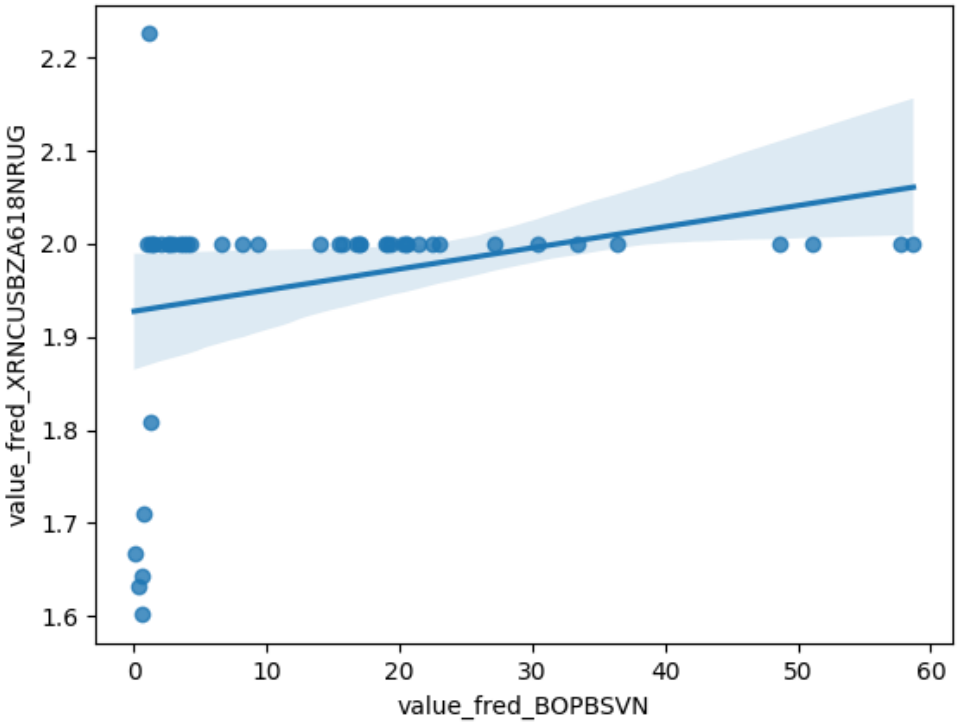


Figure 1: Raw Regression Plot for 2026-10-12

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_XRNCUSBZA618NRUG_detrended	**R-squared:**	0.123
Model:	OLS	**Adj. R-squared:**	0.103
Method:	Least Squares	**F-statistic:**	6.041
Date:	Mon, 12 Oct 2026	**Prob (F-statistic):**	0.0181
Time:	20:28:58	**Log-Likelihood:**	41.263
No. Observations:	45	**AIC:**	-78.53
Df Residuals:	43	**BIC:**	-74.91
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	5.671e-15	0.015	3.84e-13	1.000	-0.030	0.030
value_fred_BOPBSVN_detrended	-0.0050	0.002	-2.458	0.018	-0.009	-0.001

Omnibus:	8.915	**Durbin-Watson:**	0.588
Prob(Omnibus):	0.012	**Jarque-Bera (JB):**	16.952
Skew:	0.296	**Prob(JB):**	0.000208
Kurtosis:	5.948	**Cond. No.**	7.28

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

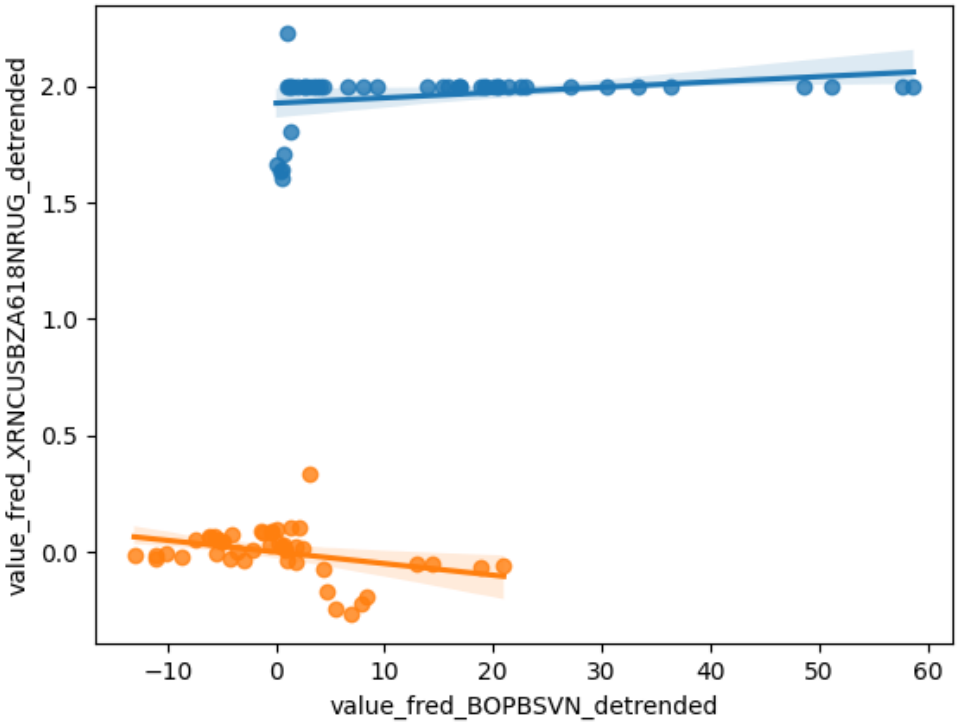


Figure 2: Detrended Regression Plot for 2026-10-12

3 Date: 2026-10-13

Series ID: APUB30074714
This series is titled Average Price: Gasoline, Unleaded Regular (Cost per Gallon/3.785 Liters) in the South Census Region - Size Class B and has a frequency of Monthly. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1978-01-01 and the observation end date is 1997-12-01.The popularity of this series is 1.

Series ID: FHA30
This series is titled 30-Year FHA Mortgage Rate: Secondary Market (DISCONTINUED) and has a frequency of Monthly. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1964-01-01 and the observation end date is 2000-06-01.The popularity of this series is 9.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_FHA30	**R-squared:**	0.169
Model:	OLS	**Adj. R-squared:**	0.165
Method:	Least Squares	**F-statistic:**	47.29
Date:	Tue, 13 Oct 2026	**Prob (F-statistic):**	5.59e-11
Time:	18:35:32	**Log-Likelihood:**	-534.79
No. Observations:	235	**AIC:**	1074.
Df Residuals:	233	**BIC:**	1081.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	3.5713	1.048	3.408	0.001	1.507	5.636
value_fred_APUB30074714	6.5135	0.947	6.876	0.000	4.647	8.380

Omnibus:	11.293	**Durbin-Watson:**	0.039
Prob(Omnibus):	0.004	**Jarque-Bera (JB):**	6.092
Skew:	0.195	**Prob(JB):**	0.0476
Kurtosis:	2.315	**Cond. No.**	13.6

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

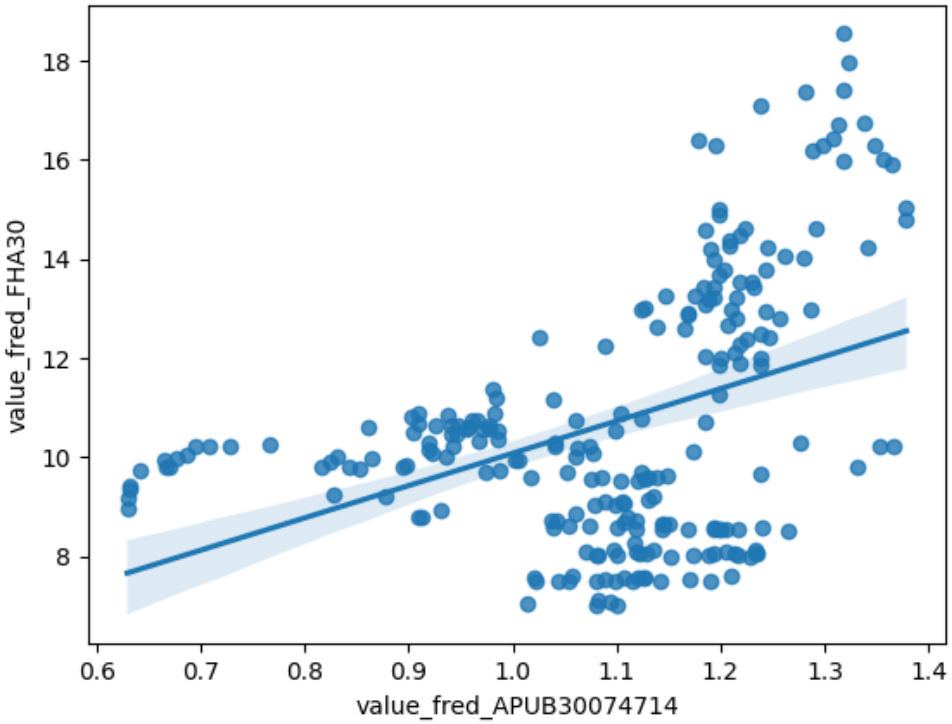


Figure 3: Raw Regression Plot for 2026-10-13

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_FHA30_detrended	**R-squared:**	0.626
Model:	OLS	**Adj. R-squared:**	0.624
Method:	Least Squares	**F-statistic:**	389.6
Date:	Tue, 13 Oct 2026	**Prob (F-statistic):**	1.23e-51
Time:	18:35:32	**Log-Likelihood:**	-352.60
No. Observations:	235	**AIC:**	709.2
Df Residuals:	233	**BIC:**	716.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	5.561e-13	0.071	7.82e-12	1.000	-0.140	0.140
value_fred_APUB30074714_detrended	8.7393	0.443	19.738	0.000	7.867	9.612

Omnibus:	5.577	**Durbin-Watson:**	0.209
Prob(Omnibus):	0.062	**Jarque-Bera (JB):**	5.236
Skew:	0.334	**Prob(JB):**	0.0729
Kurtosis:	3.299	**Cond. No.**	6.23

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

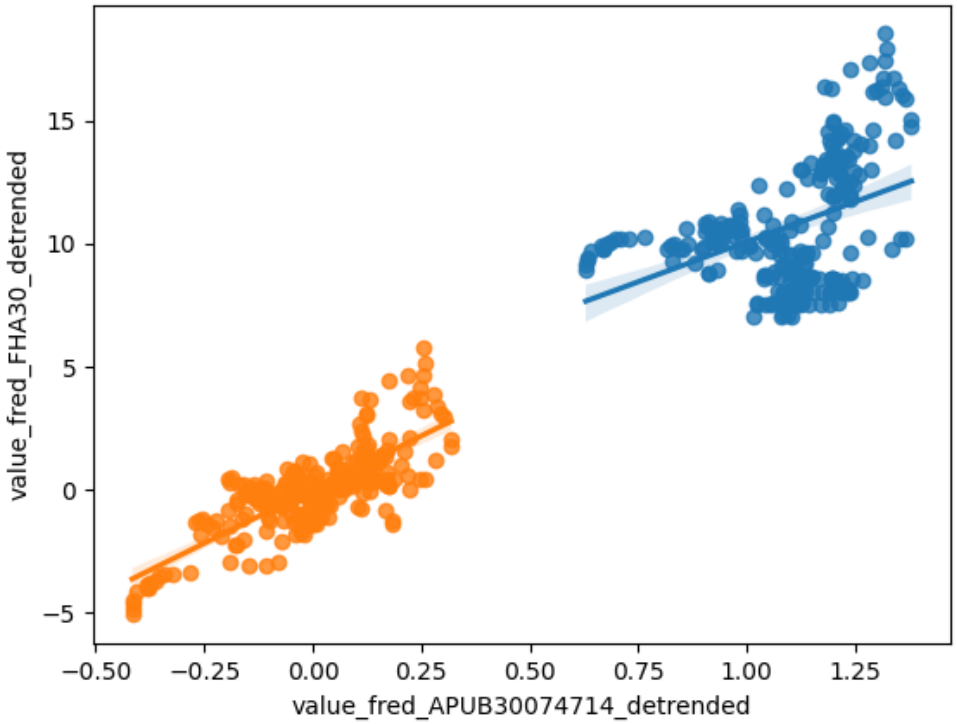


Figure 4: Detrended Regression Plot for 2026-10-13

4 Date: 2026-10-14

Series ID: QFR201333USNO
This series is titled Quarterly Financial Report: U.S. Corporations: Machinery: Cash and Demand Deposits in the U.S. and has a frequency of Quarterly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2000-10-01 and the observation end date is 2026-04-01.The popularity of this series is 1.

Series ID: GSETCMDODFS
This series is titled Government-Sponsored Enterprises; Debt Securities and Loans; Liability, Level and has a frequency of Quarterly, End of Period. The units are Millions of U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1945-10-01 and the observation end date is 2026-04-01.The popularity of this series is 2.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_GSETCMDODFS	**R-squared:**	0.805
Model:	OLS	**Adj. R-squared:**	0.803
Method:	Least Squares	**F-statistic:**	416.8
Date:	Wed, 14 Oct 2026	**Prob (F-statistic):**	1.26e-37
Time:	16:02:51	**Log-Likelihood:**	-1578.2
No. Observations:	103	**AIC:**	3160.
Df Residuals:	101	**BIC:**	3166.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.816e+05	2.74e+05	1.758	0.082	-6.18e+04	1.02e+06
value_fred_QFR201333USNO	237.3381	11.626	20.415	0.000	214.276	260.400

Omnibus:	4.636	**Durbin-Watson:**	0.398
Prob(Omnibus):	0.098	**Jarque-Bera (JB):**	4.489
Skew:	-0.511	**Prob(JB):**	0.106
Kurtosis:	2.949	**Cond. No.**	5.94e+04

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 5.94e+04. This might indicate that there are
strong multicollinearity or other numerical problems.

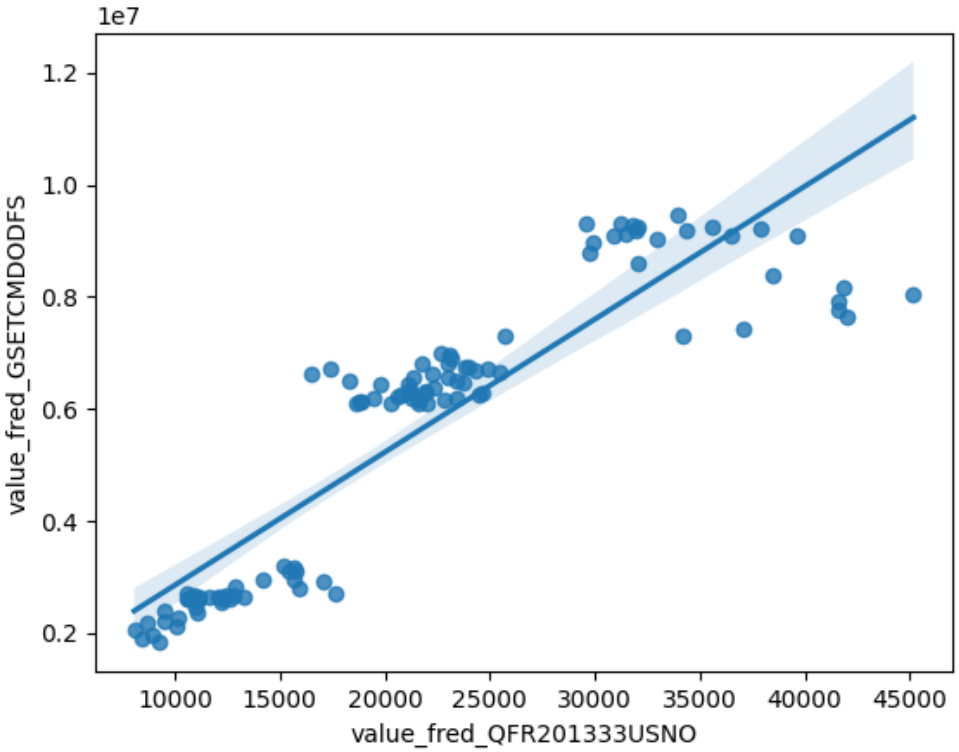


Figure 5: Raw Regression Plot for 2026-10-14

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_GSETCMDODFS_detrended	**R-squared:**	0.022
Model:	OLS	**Adj. R-squared:**	0.013
Method:	Least Squares	**F-statistic:**	2.317
Date:	Wed, 14 Oct 2026	**Prob (F-statistic):**	0.131
Time:	16:02:51	**Log-Likelihood:**	-1530.9
No. Observations:	103	**AIC:**	3066.
Df Residuals:	101	**BIC:**	3071.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-4.732e-09	6.86e+04	-6.9e-14	1.000	-1.36e+05	1.36e+05
value_fred_QFR201333USNO_detrended	28.1415	18.489	1.522	0.131	-8.536	64.819

Omnibus:	13.571	**Durbin-Watson:**	0.344
Prob(Omnibus):	0.001	**Jarque-Bera (JB):**	16.691
Skew:	0.705	**Prob(JB):**	0.000237
Kurtosis:	4.378	**Cond. No.**	3.71e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.71e+03. This might indicate that there are strong multicollinearity or other numerical problems.

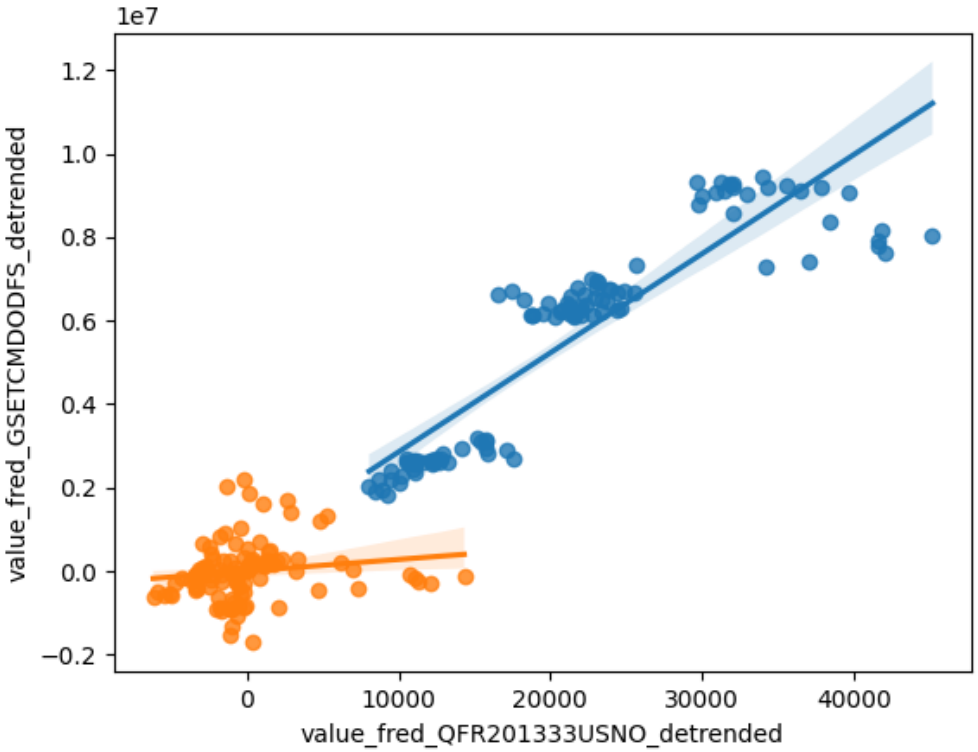


Figure 6: Detrended Regression Plot for 2026-10-14

5 Date: 2026-10-15

Series ID: PLGCPPSEA670NRUG
This series is titled Price Level of Government Consumption for Sweden and has a frequency of Annual. The units are Price Level of USA Output-side GDP in 2021=1 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1950-01-01 and the observation end date is 2023-01-01.The popularity of this series is 1.

Series ID: CUUSA214SEHF
This series is titled Consumer Price Index for All Urban Consumers: Energy Services in Kansas City, MO-KS (CMSA) (DISCONTINUED) and has a frequency of Semiannual. The units are Index 1982-1984=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1984-01-01 and the observation end date is 2017-07-01.The popularity of this series is 1.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_CUUSA214SEHF	**R-squared:**	0.326
Model:	OLS	**Adj. R-squared:**	0.305
Method:	Least Squares	**F-statistic:**	15.50
Date:	Thu, 15 Oct 2026	**Prob (F-statistic):**	0.000418
Time:	15:51:08	**Log-Likelihood:**	-162.19
No. Observations:	34	**AIC:**	328.4
Df Residuals:	32	**BIC:**	331.4
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	51.5366	22.372	2.304	0.028	5.966	97.107
value_fred_PLGCPPSEA670NRUG	84.4492	21.453	3.937	0.000	40.752	128.147

Omnibus:	0.731	**Durbin-Watson:**	0.224
Prob(Omnibus):	0.694	**Jarque-Bera (JB):**	0.812
Skew:	0.276	**Prob(JB):**	0.666
Kurtosis:	2.482	**Cond. No.**	8.76

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

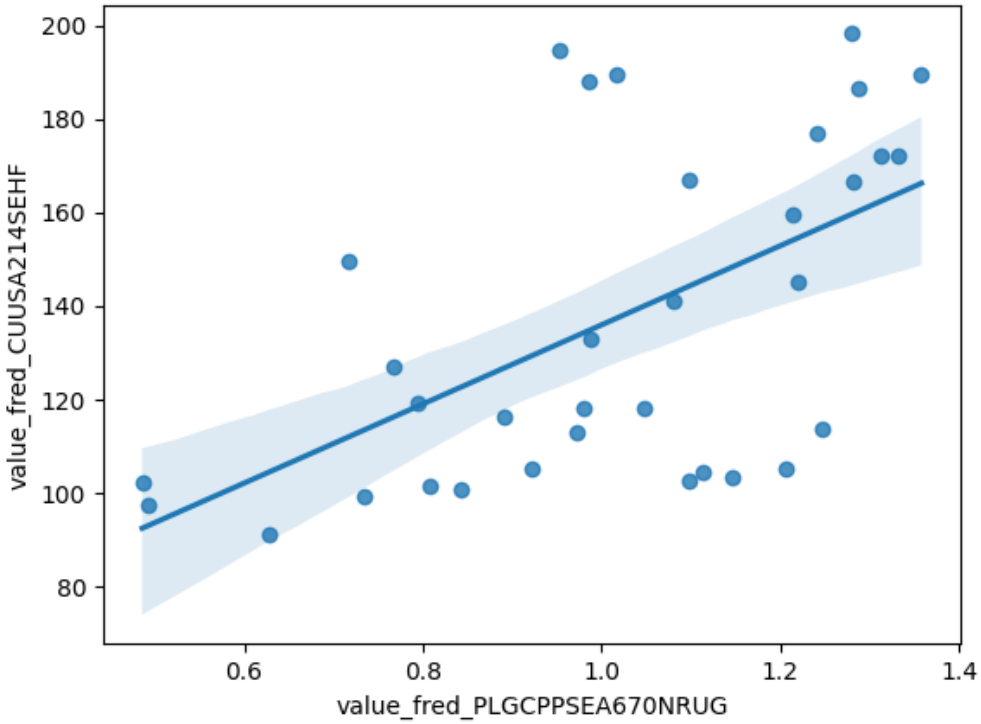


Figure 7: Raw Regression Plot for 2026-10-15

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_CUUSA214SEHF_detrended	**R-squared:**	0.006
Model:	OLS	**Adj. R-squared:**	-0.025
Method:	Least Squares	**F-statistic:**	0.1907
Date:	Thu, 15 Oct 2026	**Prob (F-statistic):**	0.665
Time:	15:51:08	**Log-Likelihood:**	-126.43
No. Observations:	34	**AIC:**	256.9
Df Residuals:	32	**BIC:**	259.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.842e-12	1.763	1.05e-12	1.000	-3.590	3.590
value_fred_PLGCPPSEA670NRUG_detrended	-4.1498	9.502	-0.437	0.665	-23.505	15.205

Omnibus:	0.585	**Durbin-Watson:**	1.001
Prob(Omnibus):	0.746	**Jarque-Bera (JB):**	0.498
Skew:	0.275	**Prob(JB):**	0.780
Kurtosis:	2.781	**Cond. No.**	5.39

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

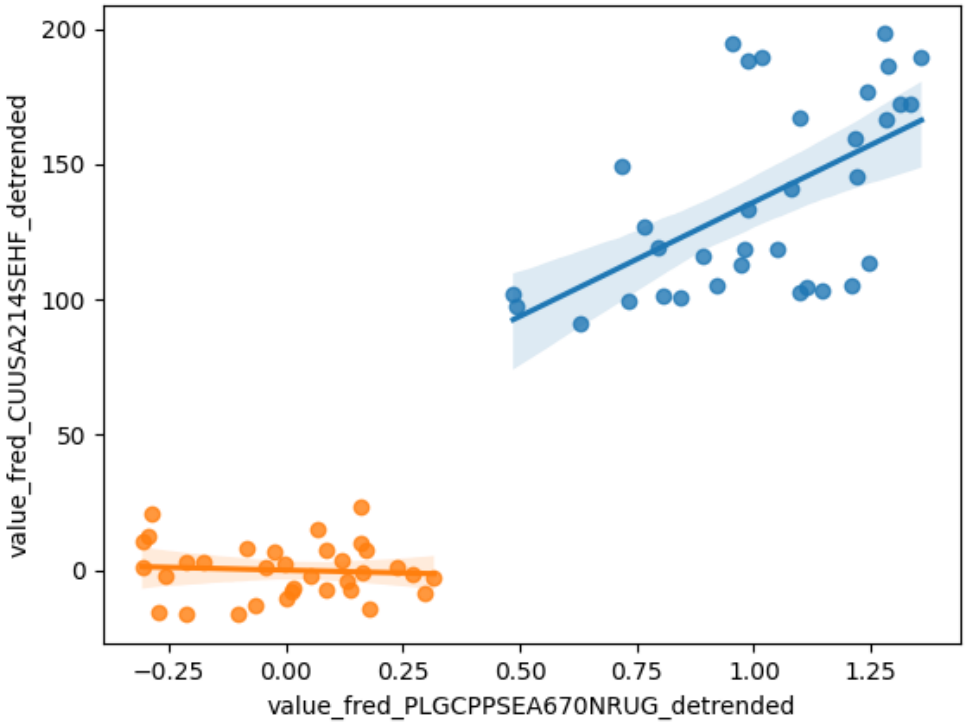


Figure 8: Detrended Regression Plot for 2026-10-15